

## ANDREW ALAGNA

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### SOFTWARE ENGINEER | FRONTEND & FULL-STACK DEVELOPMENT | ACCESSIBILITY & PERFORMANCE

Software Engineer with 3+ years building high-performance, WCAG AAA-compliant web applications at American Express. Specialized in React.js and Node.js with proven expertise in accessibility, performance optimization, and A/B testing that delivered measurable business impact for 10M+ users across 10+ international markets.

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### PROFESSIONAL EXPERIENCE

#### American Express

**Software Engineer** | 08/2022 - Present

- Spearheaded Web Content Accessibility Guidelines (WCAG) AAA compliance implementation across 10+ international markets using HTML5 and React.js with axe DevTools and screen reader testing, improving accessibility audit scores from 72% to 100%, mitigating legal risk and expanding market reach for 10M+ global users
- Engineered secure user account management system using React.js and Node.js, serving 5M+ users annually to update sensitive information (name, address, contact details, income) while maintaining data integrity and regulatory compliance
- Engineered full-stack two-factor authentication (2FA) system processing 4M+ annual updates to phone numbers and email addresses, building React.js forms with real-time validation and Kotlin-based BFF API layer to handle sensitive data updates while maintaining data integrity and regulatory compliance
- Designed and executed 16+ A/B experiments using JavaScript, analytics APIs, and statistical significance testing, including A/B testing 'Find It Fast' navigation to identify and prioritize top 3 revenue-driving links, increasing form completion rates by 12% and customer satisfaction scores by 17% across 3m+ users
- Built automated Continuous Integration/Continuous Deployment (CI/CD) pipelines using GitHub Actions, reducing deployment time by 15% and implementing feature flagging system with real-time toggles for safer releases
- Collaborated in Agile environment to deliver enterprise-scale applications, implementing a comprehensive testing strategy with Jest/React Testing Library, achieving 100% code coverage and reducing production bugs by 17%

#### CodePath

**Intro to Software Engineering - Teaching Assistant** | 05/2022 - 08/2022

- Led weekly instruction sessions for 180+ students in Python-based Data Structures & Algorithms, achieving 80% student satisfaction through clear explanations and engaging problem-solving
- Showcased 7+ live coding exercises that improved technical interview performance, mentoring students on career readiness and job search outcomes

**New York City Department of Transportation**  
**Geographic Information Systems Intern** | 04/2022 - 08/2022

- Engineered a Python program to automate census data processing and visualization, reducing processing time from weeks to seconds for citywide geospatial analysis and statistics
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**VOLUNTEER LEADERSHIP**

**CodePath** | 06/2021 - Present

- Mentored and instructed 350+ students across multiple cohorts in Data Structures & Algorithms, cybersecurity fundamentals, open source contributions, and career readiness, fostering an inclusive learning environment
- Taught technical curriculum achieving 85%+ student satisfaction and guided students through hands-on debugging and technical interview preparation, securing free certifications for 300+ students

**CUNY: Hunter College** | 06/2021 - Present

**Teaching & Mentoring**

- Providing 1-on-1 career mentorship to 2 current students on technical interviews, side project ideation, resume optimization, and navigating the tech job market through video sessions and LinkedIn messaging
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**EDUCATION**

**Bachelor of Arts in Computer Science**

Hunter College | New York, NY | Graduated: May 2022 | GPA: 3.62, Cum Laude

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**TECHNICAL SKILLS**

**Languages:** JavaScript (ES6+), TypeScript, Python, SQL, Kotlin

**Frontend:** React.js, Next.js, React Context API, Redux, HTML5, CSS3, Tailwind CSS, Responsive Design

**Backend & APIs:** Node.js, GraphQL, REST APIs, BFF Architecture, Microservices

**Tools & Practices:** Git, GitHub Actions (CI/CD), Jest, React Testing Library, WCAG AAA Accessibility, Agile/Scrum, A/B Testing, Performance Optimization, AWS, Docker